

Nathanial Cavallucci Senior AI Full Stack & Workflow Automation Engineer

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📍 Eugene, OR, US

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🐙 Github

👤 PROFILE

Builder-minded AI Automation Engineer focused on a single mission: making work disappear. I own work elimination and operational transformation, continuously looking for opportunities to simplify operations, eliminate manual effort, and redesign how work gets done. Proven expertise in intelligent workflow orchestration, process transformation, and deploying AI agents/RAG pipelines to automate repetitive approvals and eliminate manual reporting. I don't just build products people use; I build automations that remove the work people shouldn't be doing, creating capacity and driving operational efficiency across the company.

🧰 PROFESSIONAL EXPERIENCE

AI Automation & Workflow Engineer, *Simform*

05/2024 – 05/2026

Orlando, FL

- Architect and deploy AI-powered full stack applications using React, Next.js, Python, and cloud-native infrastructure across AWS and Azure environments
- Serve as Senior Enterprise Workflow Automation Specialist to migrate, build, and maintain complex K2/Nintex business process workflows in dedicated multi-month engagements
- Design production-ready RAG pipelines and AI agent systems, reducing manual data processing and approval cycle times by 60%+ for enterprise clients
- Modernize legacy BPM architectures by integrating Nintex with SharePoint, Power Automate, and custom REST/GraphQL APIs while maintaining strict governance and compliance standards
- Implement CI/CD pipelines, Docker/Kubernetes orchestration, and real-time data processing systems to scale AI and automation solutions globally

Full Stack Developer, *The Agency Engineer*

01/2025 – 02/2026

Miami Beach, FL

- Developed scalable full-stack web applications using Python, Node.js, React, Next.js, and TypeScript.
- Built and integrated REST and GraphQL APIs, third-party services, webhooks, and enterprise SaaS tools.
- Designed secure backend services and data workflows using PostgreSQL, MongoDB, Redis, and microservices patterns.
- Implemented OAuth 2.0 and JWT authentication and authorization with secure API design practices.
- Containerized applications with Docker and supported cloud deployments, CI/CD pipelines, and production troubleshooting.
- Collaborated with stakeholders and engineering teams to translate business requirements into reliable, maintainable software.

Automation Engineer & Systems Integrator, *AptlyLabsInc*

03/2022 – 12/2023

South Barrington, IL

- Developed 30+ scalable web platforms using Node.js, Python, and modern frontend frameworks, delivering secure, high-availability SaaS products
- Integrated third-party APIs, webhooks, and enterprise SaaS tools to automate cross-functional business operations and reporting workflows
- Optimized backend performance using Redis, WebSockets, and microservices architecture, improving API response times by 45%
- Collaborated with enterprise stakeholders to map manual processes to automated Nintex/K2 workflows, reducing operational overhead and improving cross-departmental visibility
- Software Engineer

Full Stack Developer, *Club Northwest*

09/2022 – 10/2023

Grants Pass, OR

- Developed and maintained full-stack web applications using Python, Node.js, React, TypeScript, and REST APIs.
- Built responsive frontend components and backend services supporting business workflows, reporting, and operational applications.
- Integrated external APIs and services to automate data exchange and streamline cross-functional business processes.
- Worked with PostgreSQL, MongoDB, and Redis to support application data, caching, and performance optimization.
- Implemented secure authentication, authorization, and API integration patterns for reliable business applications.
- Collaborated with stakeholders and QA teams through development, testing, deployment, and production support.

Software Engineer - Cloud & Data Automation, *Unstoppable You*

02/2019 – 11/2022

Miami

- Designed and maintained cloud infrastructure using AWS/Azure, Docker, Kubernetes, and Infrastructure as Code (Terraform/CloudFormation)
- Built ETL pipelines and real-time data processing systems for enterprise analytics, monitoring dashboards, and predictive modeling

- Implemented OAuth 2.0, JWT, and secure authentication frameworks for multi-tenant applications handling sensitive enterprise data
- Supported early-stage deployment of AI/ML models into production environments, focusing on scalability, monitoring, and cost optimization

 EDUCATION

B.S. in Computer Science, *University of Oregon*

- Focus: Distributed Systems, Machine Learning, Cloud Architecture, Database Engineering
- Capstone: Developed a cloud-native workflow automation prototype integrating AI decision models with enterprise BPM platforms

09/2018 – 08/2022

Eugene, OR

 SKILLS

Intelligent Automation & AI AI Agents, RAG Pipelines, LLMs, LangChain, Prompt Engineering, Vector DBs (Pinecone, Weaviate)	Workflow Orchestration & BPM K2/Nintex, Power Automate, SharePoint, Process Transformation, Legacy Workflow Migration, Business Process Optimization	Backend & Integrations Python, Node.js, REST/GraphQL APIs, Webhooks, SaaS Integrations, Microservices, Event-Driven Architecture
Cloud & Data Ops AWS, Azure, Docker, Kubernetes, Terraform, ETL/ELT Pipelines, PostgreSQL, Snowflake, Stream Processing		